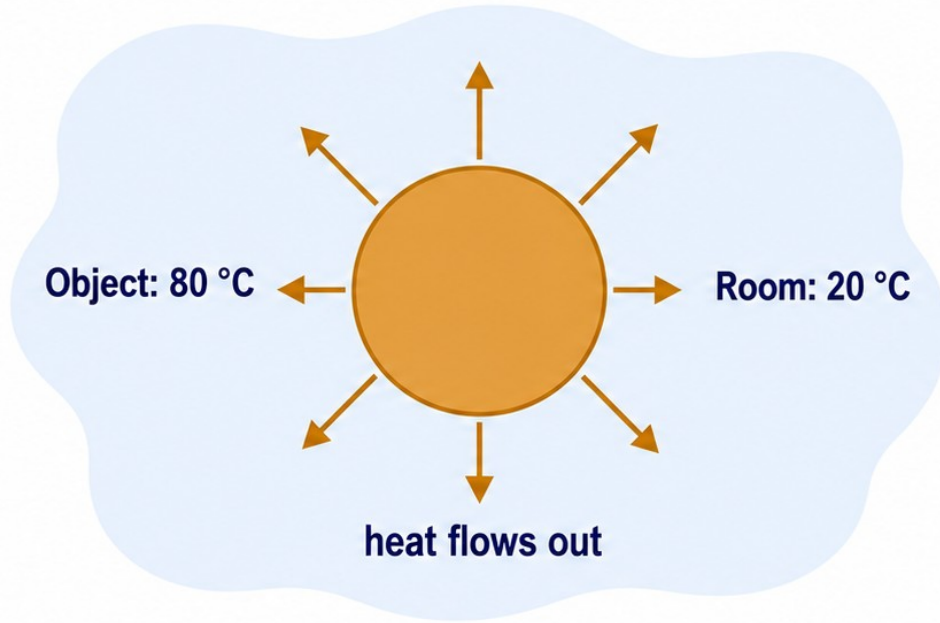


Can a Cooling Rate Predict a Temperature?



What do you predict?

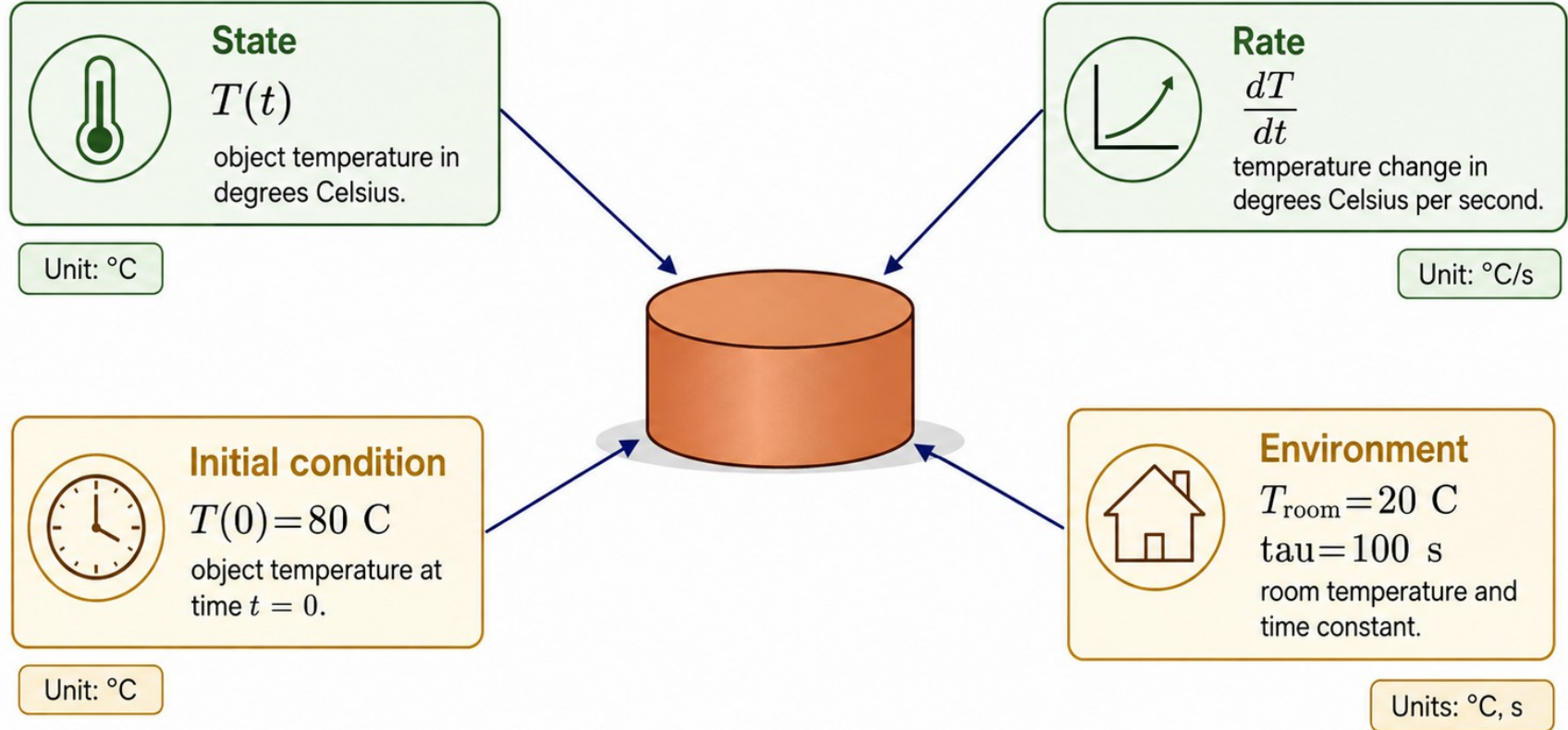
Same temperature
drop every 50 s?

Or a smaller drop
as the object cools?



Use a rate law to predict temperature from 0 to 300 s.

Name the State, Rate, and Initial Condition



Read the Cooling Law in Words

minus sign
temperature decreases
while $T > 20^\circ\text{C}$

temperature excess
larger excess $\rightarrow$
faster cooling

$$\frac{dT}{dt} = - \frac{(T - 20)}{100}$$

cooling time
100 s

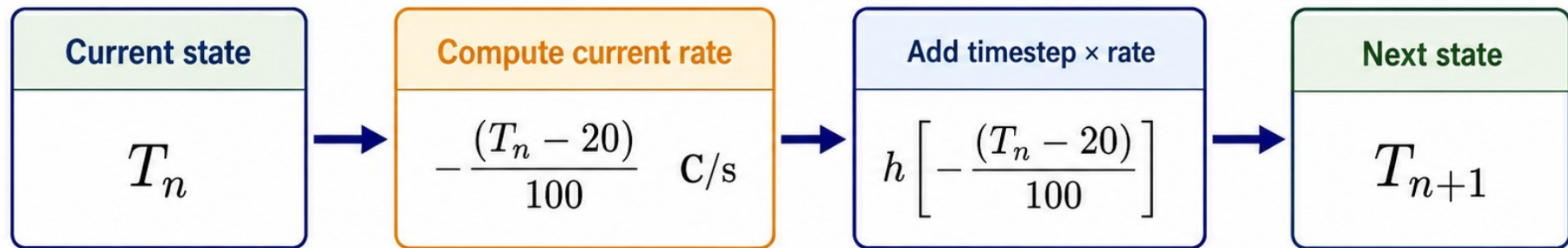
Initial rate

At $T = 80^\circ\text{C}$:

$$\frac{dT}{dt} = - \frac{(80 - 20)}{100} = -0.60^\circ\text{C/s}$$

$$\frac{\text{temperature change}}{\text{time}} = ^\circ\text{C/s}$$

Turn the Current Rate into One Step



$$T_{n+1} = T_n + h \left[-\frac{(T_n - 20)}{100} \right].$$

- Next temperature = current temperature + timestep $\times$ current rate.
- Seconds $\times$ C/s gives a temperature change.
- A smaller h recalculates the changing rate sooner.

Dimensional check:

$$h \times \text{rate: } s \times \text{C/s} = \text{C}$$

Trace Two Euler Updates

Model: $\frac{dT}{dt} = -\frac{(T - 20)}{100}$, where T is in degrees Celsius

1 Start

$t = 0$ s
 $T_0 = 80$ °C
rate = -0.60 °C/s
 $h = 50$ s

2 Update 1

$t = 50$ s
 $T_1 = 50$ °C

$$T_1 = 80 + 50(-0.60) = 50 \text{ °C}$$

3 Recalculate rate

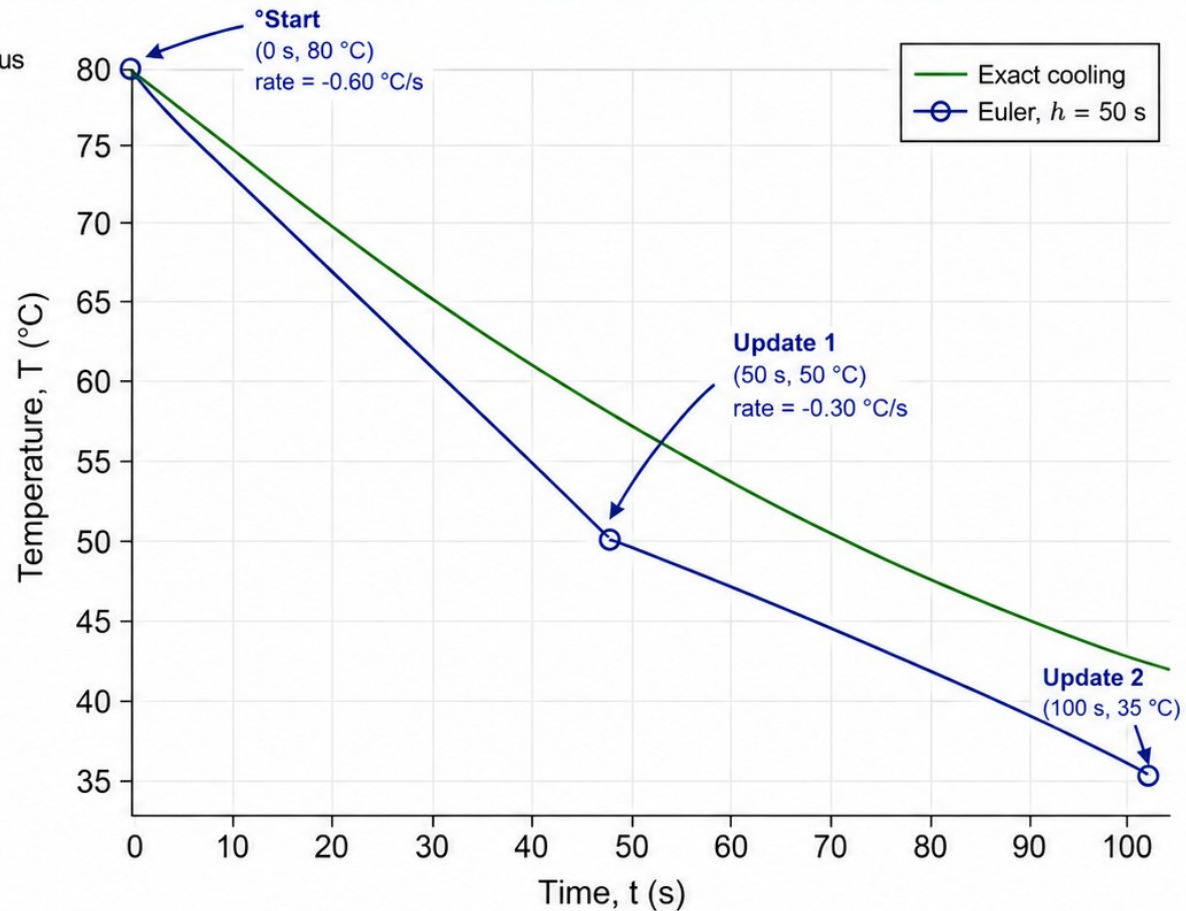
$$\text{rate} = -\frac{(50 - 20)}{100} = -0.30 \text{ °C/s}$$

4 Update 2

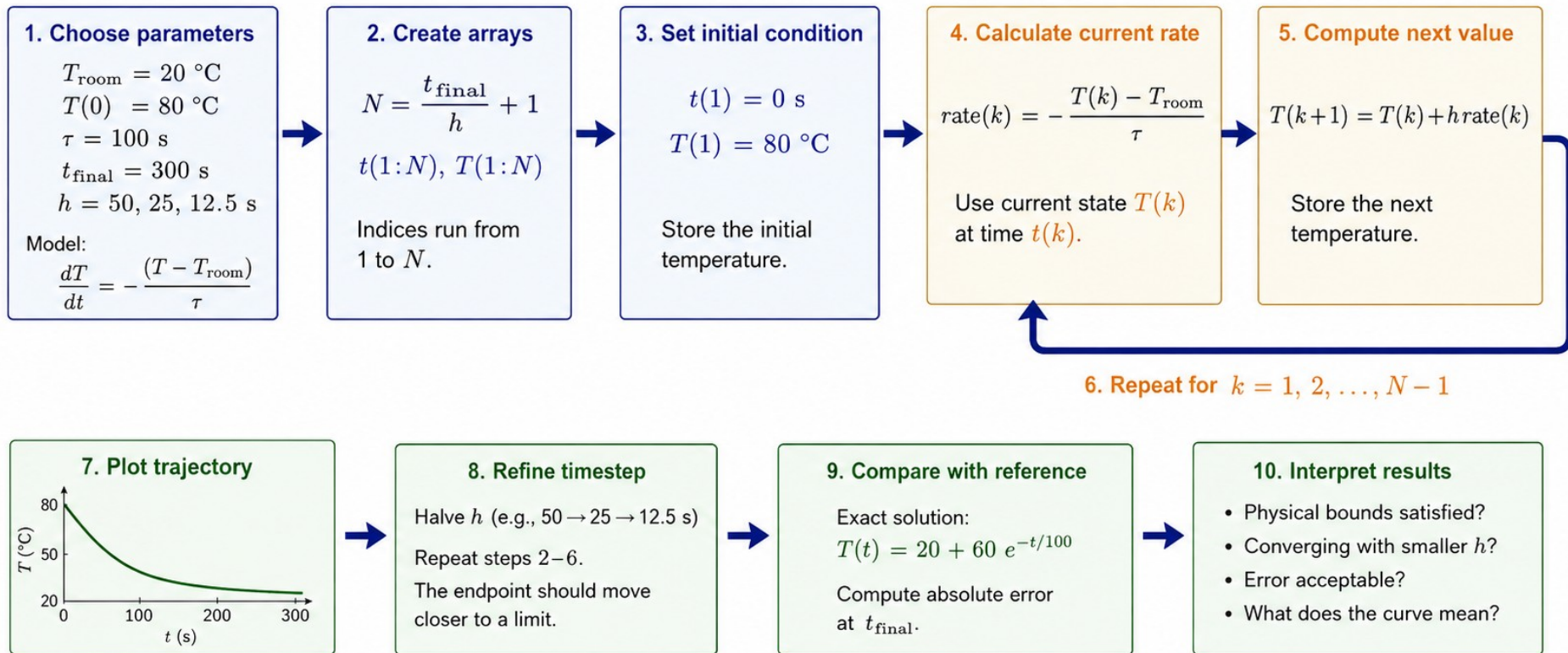
$t = 100$ s
 $T_2 = 35$ °C

$$T_2 = 50 + 50(-0.30) = 35 \text{ °C}$$

The second drop is smaller because the rate is recalculated.

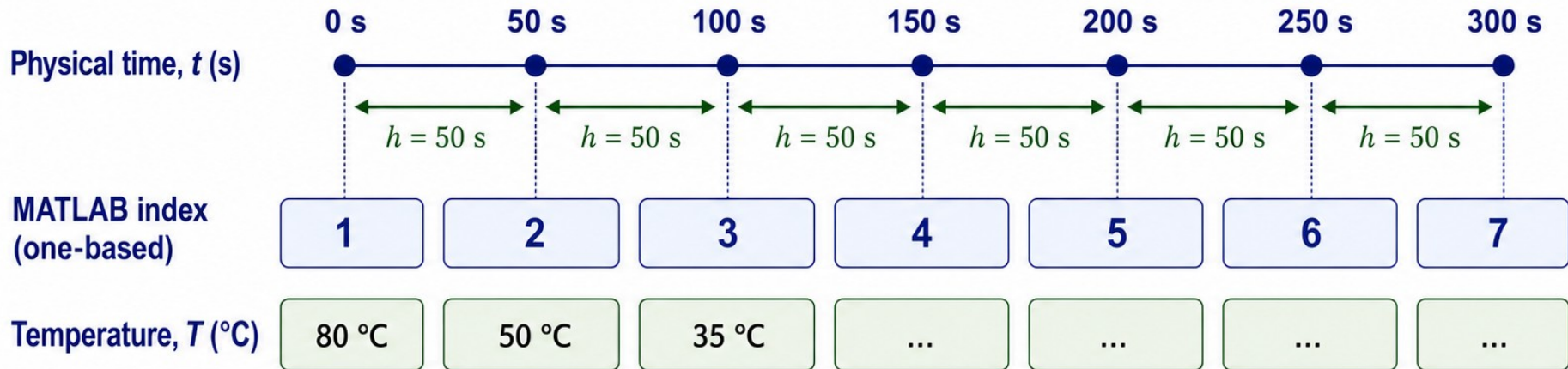


Plan the Algorithm Before MATLAB



Goal: Build the trajectory from the initial condition using repeated rate-based updates, then validate before trusting the result.

Seven Values Represent Six Intervals



MATLAB index 1 represents physical time $t = 0$ s.



Six intervals require seven stored temperatures.

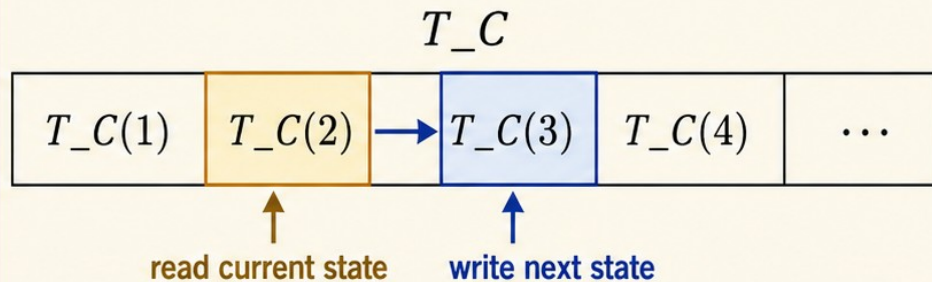
```
t_s = 0:50:300;  
T_C = zeros(size(t_s)); T_C(1) = 80;
```


Read the Euler Loop

MATLAB Code

```
for n = 1:numel(t_s)-1
    rate_Cps = -(T_C(n)-T_room_C)/tau_s;
    T_C(n+1) = T_C(n) + dt_s*rate_Cps;
end
```

Indexed Read/Write Trace

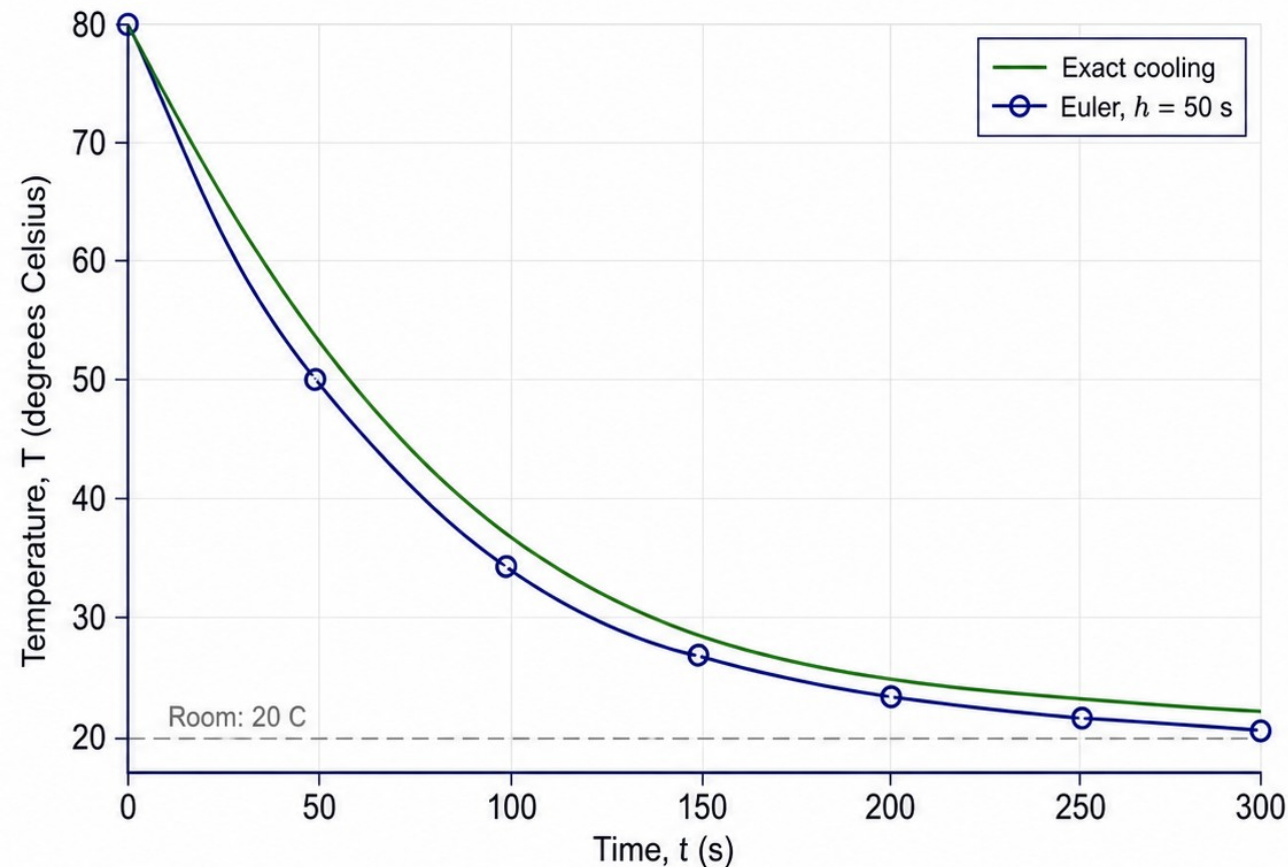


When $n = 2$, read $T_C(2)$ and write $T_C(3)$.



Stop one short: $n = 1:\text{numel}(t_s)-1$ keeps $T_C(n+1)$ inside the array.

Check the Trajectory Before Trusting It



Initial value: $T(0) = 80$ C.



The supplied trajectory decreases toward 20 C.



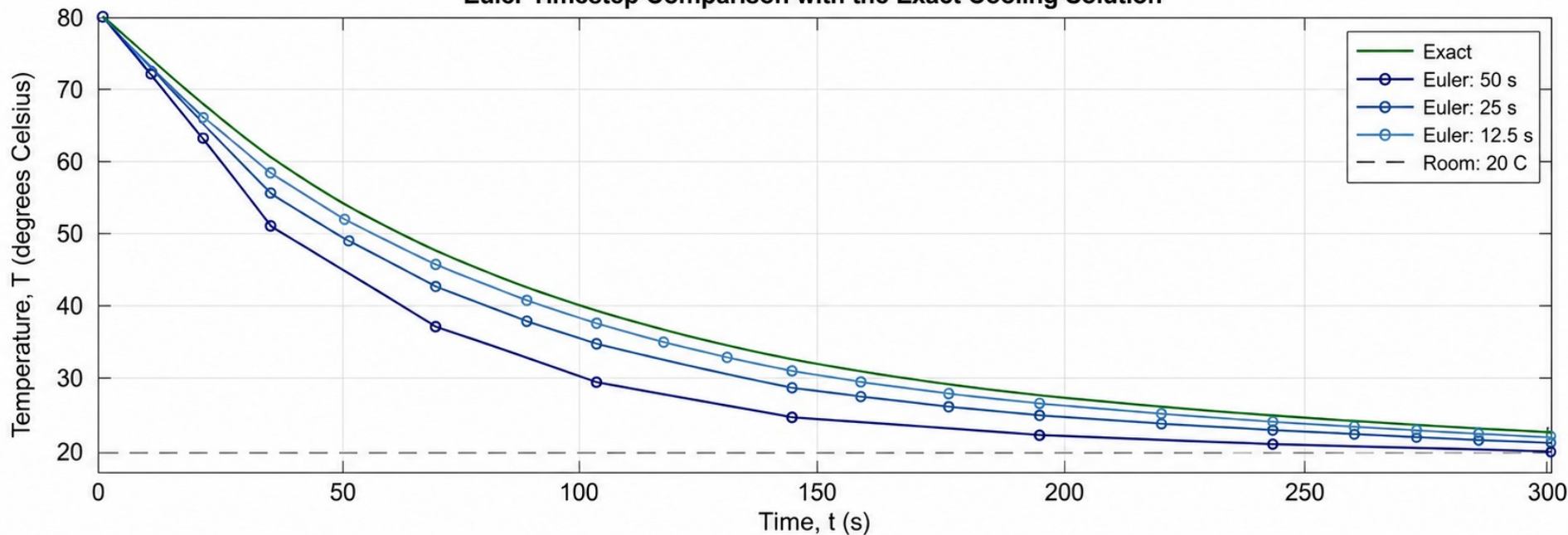
For these steps,
 $20 \text{ C} \leq T \leq 80 \text{ C}$.



A plausible trend does not establish numerical accuracy.

Change the Timestep, Keep the Physics Fixed

Euler Timestep Comparison with the Exact Cooling Solution



Fixed: $T_{room} = 20$ C



Fixed: $T(0) = 80$ C



Fixed: $\tau = 100$ s



Fixed: $t_{final} = 300$ s



Change only h : 50 s $\rightarrow$ 25 s $\rightarrow$ 12.5 s



Smaller steps
recalculate the rate sooner.



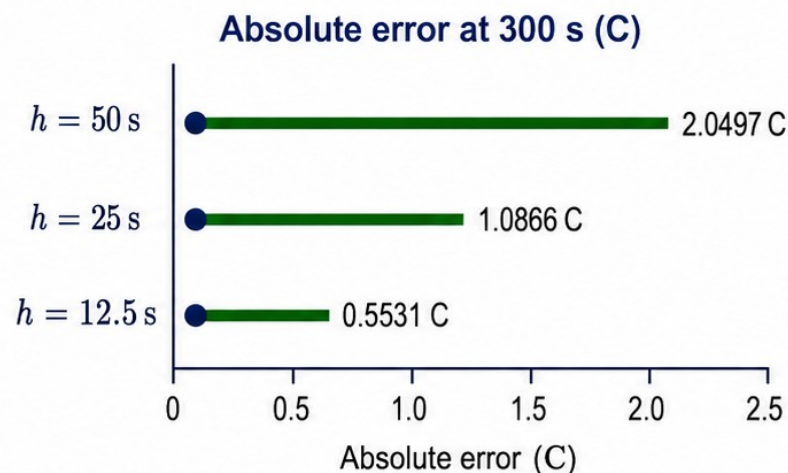
The supplied endpoints
move toward the exact curve.

Measure Error Against a Supplied Reference

$$T_{\text{exact}}(t) = 20 + 60 \exp(-t/100)$$

$$\text{Exact } T(300 \text{ s}) = 22.9872 \text{ C}$$

Timestep h (s)	Euler endpoint at 300 s (C)	Absolute error (C)
50	20.9375	2.0497
25	21.9006	1.0866
12.5	22.4341	0.5531



Use absolute error for
a temperature difference.

smaller $h \rightarrow$ smaller endpoint error

Spot a Runnable but Wrong Update

RUNS — BUT WRONG

$$T_C(n+1) = T_C(n) + \text{rate_Cps};$$

$$C + C/s$$

✗ **Unit mismatch**

Temperature plus temperature/time is dimensionally inconsistent.

REPAIRED FORWARD EULER UPDATE

$$T_C(n+1) = T_C(n) + dt_s * \text{rate_Cps};$$

$$C + s(C/s) = C$$

✓ **Units agree**



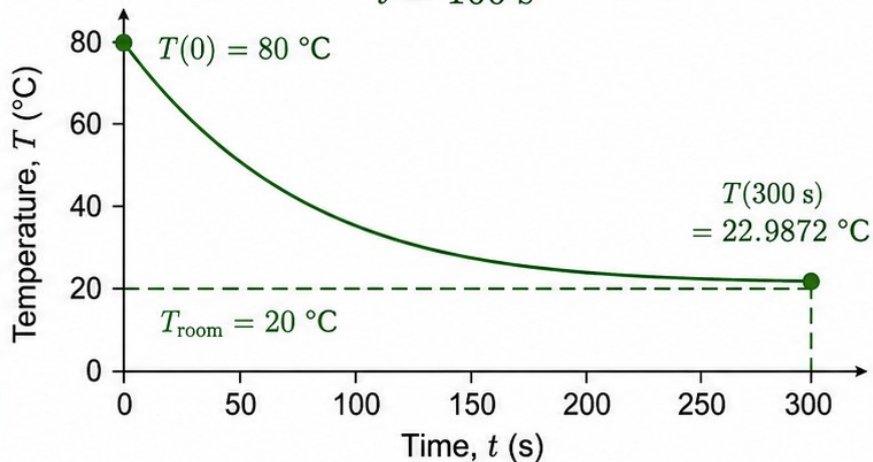
Diagnostic: Where does the timestep belong?

A smooth-looking graph is not a validation check.

Predict a Slower Cooling Model

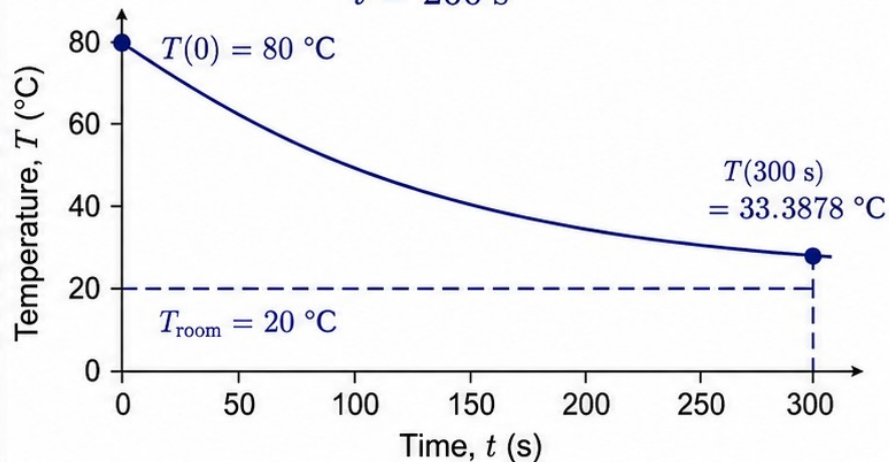
Faster physical cooling

$$\tau = 100 \text{ s}$$



Slower physical cooling

$$\tau = 200 \text{ s}$$



τ is physical

Change τ : cooling rate changes
initial $|rate|$: $0.60 \rightarrow 0.30^{\circ}\text{C/s}$



h is numerical

Change h : accuracy changes,
not the physical model



A smaller timestep cannot fix a nonuniform-temperature or changing-room model limitation.

Explain One Update and One Check

1



1. Identify the model

Name the state, rate unit, and initial condition.

State: T (degrees Celsius)

Rate: degrees Celsius per second

Initial condition: $T(0) = 80$ C

2



2. Make one update

With $h = 25$ s, calculate the first Euler temperature: 65 C.

$$80 + 25(-0.60) = 65 \text{ C}$$

3



3. Explain the repeated step

Explain why the rate must be recalculated after each step.

The rate depends on the current temperature.

4



4. Choose evidence

Choose one physical check and explain why a reference comparison adds evidence.

physical bound

exact reference

State

Define the model and starting values.



Update

Use the current rate to find the next value.



Check

Apply a physical check and/or compare to exact.



Interpretation

Decide what the result means and next steps.